LSTM

W zadaniach NLP (Natural Language Processing), takich jak:

- analiza sentymentu,
- tłumaczenie maszynowe,
- generowanie tekstu,
- predykcja kolejnego słowa, potrzebujemy modeli, które rozumieją sekwencję słów, a nie tylko pojedyncze słowa.

Tu właśnie używamy **LSTM** — sieci, która potrafi pamiętać kontekst w czasie (np. zależności gramatyczne lub semantyczne).

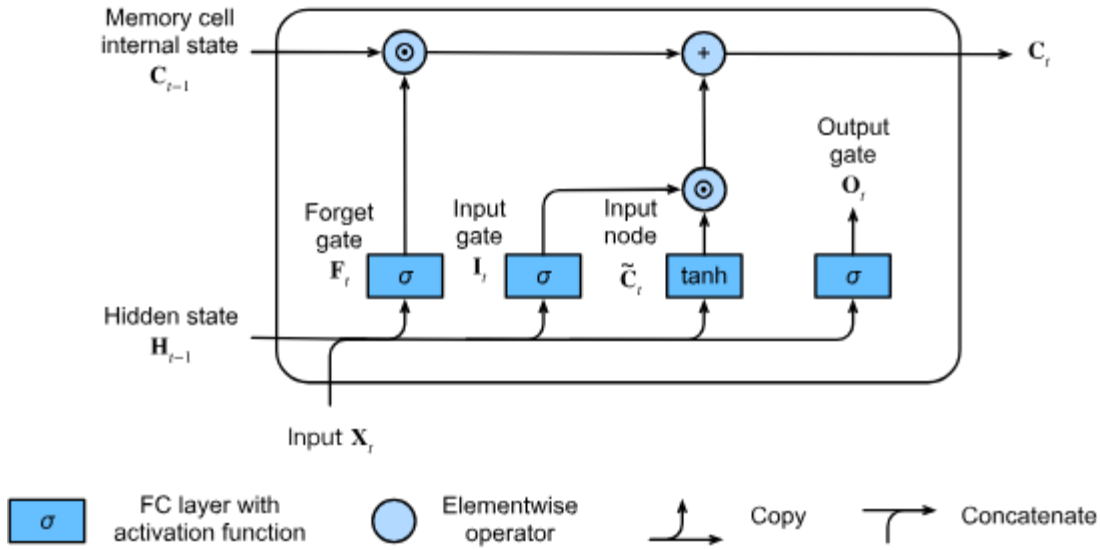
Kiedy używać LSTM?

- gdy sekwencje mają od umiarkowanej do dużej długości i **zależności czasowe są ważne**
- gdy mamy mało danych i nie chcemy używać cięższych modeli (np. transformery - to już niedługo :)), albo chcemy modelu, który działa dobrze na mniejszych zbiorach
- do zadań sekwencyjnych

Kiedy nie używać?

- gdy mamy ogromne zbiory i potrzebujemy najlepszych wyników — transformery (np. BERT) często biją LSTM
- gdy zadanie wymaga globalnych zależności i kontekstów, które lepiej łapią mechanizmy self-attention

Jak zbudowany jest LSTM?



Podstawowy blok LSTM składa się z: stanu komórkowego (cell state) i wewnętrznych bramek:

- **forget gate** (bramka zapominania) — decyduje, które informacje z poprzedniej komórki zignorować
- **input gate** (bramka wejścia) — decyduje, które nowe informacje dodać
- **output gate** (bramka wyjścia) — decyduje, jaka część stanu komórkowego stanie się wyjściem

Przykład - przewidywanie kolejnego słowa w zdaniu

In [175…
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!pip install flair
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Requirement already satisfied: accelerate>=0.26.0 in /home/damian/szkola/PJN_
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Requirement already satisfied: psutil in /home/damian/szkola/PJN_Laboratoria/
venv/lib64/python3.12/site-packages (from accelerate>=0.26.0->transformers<
5.0.0,>=4.25.0->transformers[sentencepiece]<5.0.0,>=4.25.0->flair) (7.2.2)
Requirement already satisfied: click==8.5.0 in /home/damian/szkola/PJN_Labora
toria/venv/lib64/python3.12/site-packages (from wikipedia-api>=0.5.7->flair)
(8.5.0)
Requirement already satisfied: httpx==0.28.1 in /home/damian/szkola/PJN_Labor
atoria/venv/lib64/python3.12/site-packages (from wikipedia-api>=0.5.7->flair)
(0.28.1)
Requirement already satisfied: tenacity==9.1.4 in /home/damian/szkola/PJN_Lab
oratoria/venv/lib64/python3.12/site-packages (from wikipedia-api>=0.5.7->flai
r) (9.1.4)
Requirement already satisfied: anyio in /home/damian/szkola/PJN_Laboratoria/v
env/lib64/python3.12/site-packages (from httpx==0.28.1->wikipedia-api>=0.5.7-
>flair) (4.15.1)
Requirement already satisfied: certifi in /home/damian/szkola/PJN_Laboratori
a/venv/lib64/python3.12/site-packages (from httpx==0.28.1->wikipedia-api>=
0.5.7->flair) (2026.10.19)
Requirement already satisfied: httpcore==1.* in /home/damian/szkola/PJN_Labor
atoria/venv/lib64/python3.12/site-packages (from httpx==0.28.1->wikipedia-api
>=0.5.7->flair) (1.0.9)
Requirement already satisfied: idna in /home/damian/szkola/PJN_Laboratoria/ve
nv/lib64/python3.12/site-packages (from httpx==0.28.1->wikipedia-api>=0.5.7->
flair) (3.20)
Requirement already satisfied: h11>=0.16 in /home/damian/szkola/PJN_Laborator
ia/venv/lib64/python3.12/site-packages (from httpcore==1.*->httpx==0.28.1->wi
kipedia-api>=0.5.7->flair) (0.16.0)
Requirement already satisfied: soupsieve>=1.6.1 in /home/damian/szkola/PJN_La
boratoria/venv/lib64/python3.12/site-packages (from beautifulsoup4->gdown>=
4.4.0->flair) (3.0.3)
Requirement already satisfied: sortedcontainers in /home/damian/szkola/PJN_La
boratoria/venv/lib64/python3.12/site-packages (from intervaltree->bioc<
3.0.0,>=2.0.0->flair) (2.4.0)
Requirement already satisfied: MarkupSafe>=2.0 in /home/damian/szkola/PJN_Lab
oratoria/venv/lib64/python3.12/site-packages (from jinja2->mpld3>=0.3->flair)
(3.0.4)
```

```
Requirement already satisfied: charset_normalizer<4,>=2 in /home/damian/szkol
a/PJN_Laboratoria/venv/lib64/python3.12/site-packages (from requests->transfo
rmers<5.0.0,>=4.25.0->transformers[sentencepiece]<5.0.0,>=4.25.0->flair)
(3.5.2)
Requirement already satisfied: PySocks!=1.5.7,>=1.5.6 in /home/damian/szkola/
PJN_Laboratoria/venv/lib64/python3.12/site-packages (from requests[socks]->gd
own>=4.4.0->flair) (1.7.1)

[notice] A new release of pip is available: 26.0.1 -> 26.2.2
[notice] To update, run: pip install --upgrade pip
```

In [176…
```python
import torch
import torch.nn as nn
import torch.optim as optim
from torch.utils.data import Dataset, DataLoader
import numpy as np
import random

from flair.data import Sentence
from flair.embeddings import FlairEmbeddings
```

In [177…
```python
SEED = 42
random.seed(SEED)
torch.manual_seed(SEED)
```

Out[177… `<torch._C.Generator at 0x7f87e2512ad0>`

Tworzymy słownik

In [178…
```python
sentences = [
  "It was a cold morning in October.",
  "Anna stood by the window, watching the fog rise over the city.",
  "She wrapped her scarf tighter and checked her phone.",
  "A message from Tom appeared: 'Are you ready for today?'.",
  "She smiled, remembering their plan.",
  "They were going to meet at the train station at dawn.",
  "The streets were silent, except for the sound of her footsteps.",
  "As she turned the corner, she saw a figure waiting in the mist.",
  "It was him."
]

# Inicjalizacja embeddingów
embedding = FlairEmbeddings('news-forward')

# dla każdego zdania tworzymy obiekty Sentence
flair_sentences = [Sentence(s) for s in sentences]

# Wydobycie embeddingów słów
for s in flair_sentences:
    embedding.embed(s)

# Zbudowanie słownika słów na podstawie korpusu
words = set(" ".join(sentences).split())
word2idx = {word: i+1 for i, word in enumerate(words)}  # indeksy od 1 (0 =
```

```
idx2word = {i: word for word, i in word2idx.items()}
vocab_size = len(word2idx) + 1

print("Słownik:", word2idx)
print("Rozmiar słownika:", vocab_size)
```

```
Słownik: {'silent,': 1, 'were': 2, 'watching': 3, 'and': 4, 'message': 5, 'at
': 6, 'a': 7, 'smiled,': 8, 'except': 9, 'going': 10, 'She': 11, 'Tom': 12, '
October.': 13, 'of': 14, 'Anna': 15, 'was': 16, 'city.': 17, 'wrapped': 18, '
stood': 19, "'Are": 20, 'A': 21, 'from': 22, 'fog': 23, 'for': 24, 'morning':
25, 'cold': 26, 'ready': 27, 'They': 28, 'meet': 29, 'footsteps.': 30, 'waiti
ng': 31, 'remembering': 32, 'him.': 33, 'It': 34, 'window,': 35, 'tighter': 3
6, 'you': 37, 'she': 38, 'their': 39, "today?'.": 40, 'turned': 41, 'in': 42,
'to': 43, 'figure': 44, 'rise': 45, 'scarf': 46, 'appeared:': 47, 'the': 48,
'train': 49, 'streets': 50, 'plan.': 51, 'phone.': 52, 'over': 53, 'dawn.': 5
4, 'saw': 55, 'by': 56, 'As': 57, 'The': 58, 'mist.': 59, 'checked': 60, 'sou
nd': 61, 'station': 62, 'corner,': 63, 'her': 64}
Rozmiar słownika: 65
```

Każde słowo dostaje unikalny numer.

Indeks 0 zarezerwowany na padding (wyrównywanie długości sekwencji).

Przewidywanie kolejnego słowa

Będzi trenować lstm w kontekście całej sekwencji, żeby przewidywał nam kolejne słowo.

Np: "To był chłodny" -> "poranek"

Tworzymy dataset na podstawie nazszych zdań

In [179…

```
import torch
import torch.nn as nn
import torch.optim as optim


all_words = " ".join(sentences).split() # wszystkie słowa (to co mamy w słow
seq_length = 4 # będziemy przewidywać piąte słowo na podstawie 4 poprzednich

print(len(all_words))

X_data, y_data = [], []
# sztucznie tworzymy przykłady
for i in range(len(all_words) - seq_length):
  # przesuwamy nasze okienko sekwencji słów po tekście
  X_seq = all_words[i:i+seq_length]
  y_seq = all_words[i+seq_length]
  # nie używamy słów tylko index ze slownika
  X_data.append([word2idx[w] for w in X_seq])
  y_data.append(word2idx[y_seq])


# robimy tensory na których będziemy trenować
```

```
X = torch.tensor(X_data)
y = torch.tensor(y_data)

print(f'X size: {X.size()}\n{X[:5]}\ny size:{y.size()}\n{y}')

print(f"Sample input: {X_data[0]} -> target: {y_data[0]}")
```

```
81
X size: torch.Size([77, 4])
tensor([[34, 16,  7, 26],
        [16,  7, 26, 25],
        [ 7, 26, 25, 42],
        [26, 25, 42, 13],
        [25, 42, 13, 15]])
y size:torch.Size([77])
tensor([25, 42, 13, 15, 19, 56, 48, 35,  3, 48, 23, 45, 53, 48, 17, 11, 18, 6
4,
        46, 36,  4, 60, 64, 52, 21,  5, 22, 12, 47, 20, 37, 27, 24, 40, 11,
8,
        32, 39, 51, 28,  2, 10, 43, 29,  6, 48, 49, 62,  6, 54, 58, 50,  2,
1,
         9, 24, 48, 61, 14, 64, 30, 57, 38, 41, 48, 63, 38, 55,  7, 44, 31, 4
2,
        48, 59, 34, 16, 33])
Sample input: [34, 16, 7, 26] -> target: 25
```

Na co patrzymy?

Przygotowaliśmy sobie mechanizm że model dostaje 4 słowa (o indeksach 45, 20, 13, 17), a jego zadaniem jest przewidzieć kolejne słowo (o indeksie 5).

Stwórzmy model

Stwórzmy model który od razu zrobi to co ręcznie zrobiliśmy u góry

In [180…

```
class WordPredictorLSTM(nn.Module):
  def __init__(self, vocab_size, embedding_dim, hidden_dim):
    super().__init__()
    # to nasz embeding modelu - go też będziemy trenować!
    # każde słowo zamieniamy na wektor o długości embedding_dim
    self.embedding = nn.Embedding(vocab_size, embedding_dim)
    # tu jest nasz model :)
    self.lstm = nn.LSTM(embedding_dim, hidden_dim, batch_first=True)
    # mapujemy wynik LSTM na rozkład prawdopodobieństwa z wszystkich słów w
    self.fc = nn.Linear(hidden_dim, vocab_size)


  def forward(self, x):
    # x - przyszło do nas słowo (index w słowniku)
    # zmieniamy index na wektor
    embeds = self.embedding(x)
    # przepuszczamy przez lstm
    # out: ukryte stany dla każdego kroku czasowego,
    # h: ostatni ukryty stan (zawiera podsumowanie sekwencji!) - Ważne !,
```

```
    # c: wewnętrzny stan pamięci.
    out, (h, c) = self.lstm(embeds)
    # ostatni ukryty stan, zwracamy prawdopodobieństwa - które słowo ma najw
    return self.fc(h[-1])


model_word = WordPredictorLSTM(vocab_size=vocab_size, embedding_dim=100, hid
criterion = nn.CrossEntropyLoss()
optimizer = optim.Adam(model_word.parameters(), lr=0.001)
```

Trening

In [181…

```
EPOCHS = 200
for epoch in range(EPOCHS):
  optimizer.zero_grad() # czyścimy gradienty
  output = model_word(X) # przewidywania modelu
  loss = criterion(output, y) # porównujemy przewidywanie vs rzeczywiste sło
  loss.backward() # liczymy gradienty
  optimizer.step() # aktualizujemy wagi
  if (epoch+1) % 40 == 0:
    print(f"Epoch {epoch+1}/{EPOCHS}, Loss: {loss.item():.4f}")
```

```
Epoch 40/200, Loss: 1.5044
Epoch 80/200, Loss: 0.0848
Epoch 120/200, Loss: 0.0282
Epoch 160/200, Loss: 0.0160
Epoch 200/200, Loss: 0.0106
```

Przetestujmy nasz model! 🐱

In [182…

```
def generate_text(model, seed_text, next_words=8):
  model.eval() # nie aktualizuj wag
  words = seed_text.split() # pierwsze zdanie zamieniamy na listę słów
  for _ in range(next_words): # generujemy tyle kolejnych słów ile chcemy
    seq = [word2idx.get(w, 0) for w in words[-seq_length:]] # zmieniamy na i
    x = torch.tensor(seq).unsqueeze(0) # tensor o wymiarze (1, długość konte
    with torch.no_grad():
      output = model(x) # przewidujemy na podstawie otrzymanych 4 słów
      predicted_idx = torch.argmax(output, dim=1).item() # wybieramy jakie s
      predicted_word = idx2word.get(predicted_idx, '') # indext wybranego sł
      words.append(predicted_word)
  return ' '.join(words) # zwracamy wszystko po kolei (nasza historia)


seed = "It was a cold"
print("Generated continuation:")
print(generate_text(model_word, seed))
```

```
Generated continuation:
It was a cold morning in October. Anna stood by the window,
```

Zadanie

3 pkt

Stwórz sieć LSTM, która przewidzi kolejne zdanie a nie kolejne słowo

Input sentence → Encoder LSTM → Hidden state → Decoder LSTM → Predicted next sentence

UWAGA! Zproponowano szkielet LSTM SeqToSeq - ale jeśli chcesz zrobić to w inny sposób, użyć innych emebedingów itp - śmiało :)

In [183…

```
sentences = [
    "It was a bright and sunny morning.",
    "The birds chirped happily in the trees.",
    "A cool breeze swept through the streets.",
    "Children played joyfully in the park.",
    "The cat lounged lazily on the windowsill.",
    "The coffee shop smelled of freshly brewed espresso.",
    "Raindrops pattered softly against the roof.",
    "The sky turned shades of pink and orange at sunset.",
    "A dog barked loudly at a passing car.",
    "The leaves rustled under the feet of walkers.",
    "Smoke rose from the chimney into the crisp air.",
    "The waves crashed rhythmically onto the shore.",
    "A lone cyclist rode along the quiet road.",
    "Flowers bloomed vibrantly in the garden.",
    "The old bookstore had a comforting aroma of paper.",
    "Fog rolled over the hills in the early morning.",
    "A gentle hum of traffic filled the city streets.",
    "The clock struck noon with a resounding chime.",
    "Windows rattled in the wind of the approaching storm.",
    "The bakery displayed an array of fresh pastries.",
    "A distant train whistle echoed through the valley.",
    "Stars twinkled faintly in the darkening sky.",
    "The playground was empty as dusk approached.",
    "A fisherman cast his line into the calm river.",
    "The scent of pine trees lingered in the air.",
    "Clouds gathered ominously over the mountains.",
    "The market buzzed with people and chatter.",
    "A child laughed as a balloon floated away.",
    "The lamp post flickered in the evening gloom.",
    "Shadows stretched long across the empty street.",
    "The fountain gurgled peacefully in the square.",
    "A paper airplane sailed across the classroom.",
    "The dog chased its tail in playful circles.",
    "The train station was crowded with travelers.",
    "Snowflakes drifted silently from the gray sky.",
    "A painter worked diligently on his canvas.",
    "The wind whispered secrets through the trees.",
    "A street musician played a soulful tune.",
    "The river sparkled under the midday sun.",
    "The fire crackled warmly in the hearth.",
    "A cyclist narrowly avoided a puddle on the road.",
    "The library was silent except for turning pages.",
    "The airplane soared high above the clouds.",
```

```
    "A butterfly landed delicately on a flower.",
    "The aroma of freshly baked bread filled the kitchen.",
    "The old clock ticked steadily in the hallway.",
    "A cat chased a shadow across the wall.",
    "The garden gate creaked in the evening breeze.",
    "The moon reflected off the still lake.",
    "A boy flew a kite on the open field.",
    "The streets glistened after a sudden rain shower."
]
```

In [184…

```
# EMBEDINGI
from flair.embeddings import DocumentPoolEmbeddings

sentence_embeddings = DocumentPoolEmbeddings([embedding])
flair_sentences = [Sentence(s) for s in sentences]
for s in flair_sentences:
    sentence_embeddings.embed(s)

# jeśli tu jest zero to prawdopodobnie coś poszło nie tak :)
print("Embedding pierwszego zdania:", flair_sentences[0].embedding.shape)

# DATASET - sequence to sequence
X_seq = torch.stack([flair_sentences[i].embedding.detach().cpu() for i in ra
y_seq = torch.stack([flair_sentences[i+1].embedding.detach().cpu() for i in

X_seq = X_seq.unsqueeze(1)
y_seq = y_seq.unsqueeze(1)

# jeśli tu jest zero to prawdopodobnie coś poszło nie tak :)
print("X_seq wymiar:", X_seq.shape)

# MODEL
class Seq2SeqLSTM(nn.Module):
    def __init__(self, input_dim, hidden_dim):
        super().__init__()
        self.encoder = nn.LSTM(input_dim, hidden_dim, batch_first=True)# TOD
        self.decoder = nn.LSTM(hidden_dim, hidden_dim, batch_first=True) # T
        # TODO - coś więcej? :)
        self.fc = nn.Linear(hidden_dim, input_dim)

    def forward(self, x):
        # TODO
        # encoder
        en_out, (en_h, en_c) = self.encoder(x)
        # decoder
        de_out, (de_h, de_c) = self.decoder(en_out, (en_h, en_c))
        # return
        return self.fc(de_out)

input_dim = X_seq.shape[2]
hidden_dim = 64
model_seq = Seq2SeqLSTM(input_dim, hidden_dim)

optimizer = optim.Adam(model_seq.parameters(), lr=0.001)

EPOCHS = 200
```

```
for epoch in range(EPOCHS):
    optimizer.zero_grad()
    output = model_seq(X_seq)
    loss = 1 - F.cosine_similarity(output, y_seq, dim=-1).mean()
    loss.backward()
    optimizer.step()
    if (epoch + 1) % 40 == 0:
        print(f"Epoch {epoch+1}/{EPOCHS}, Loss: {loss.item():.4f}")
```

```
Embedding pierwszego zdania: torch.Size([2048])
X_seq wymiar: torch.Size([50, 1, 2048])
Epoch 40/200, Loss: 0.2438
Epoch 80/200, Loss: 0.1826
Epoch 120/200, Loss: 0.1706
Epoch 160/200, Loss: 0.1440
Epoch 200/200, Loss: 0.1148
```

UWAGA: **Twój model nie jest modelem generatywnym**! Twój LSTM przewiduje embedding następnego zdania, a my znajdziemy w zbiorze zdań to, które jest najbardziej podobne.

In [185…

```
import torch
import torch.nn.functional as F

# Podajemy zdanie startowe

X_input = torch.tensor(flair_sentences[-3].embedding.cpu().detach().numpy())

# Model w trybie ewaluacyjnym
model_seq.eval()
with torch.no_grad():
    predicted_embedding = model_seq(X_input)

print("Wymiary przewidzianego embedingu", predicted_embedding.shape)

# Zamieniamy na 1D tensor
pred_emb = predicted_embedding.squeeze(0).squeeze(0)
# dużo problemów w AI to niezgadzające się shape'y - PAMIĘTAJ PRINTOWAĆ ROZM

# Tworzymy listę embeddingów wszystkich zdań w zbiorze
emb_list = [s.embedding.cpu() for s in flair_sentences]
# Obliczamy podobieństwo cosinusowe do wszystkich zdań
similarities = [F.cosine_similarity(pred_emb, e, dim=0) for e in emb_list]
# Wybieramy indeks najbardziej podobnego zdania
best_idx = torch.argmax(torch.tensor(similarities))
# wygląda jak niedawne zadanie z labów z wektorami słów, prawda? :)

print('Początek historii: ', flair_sentences[-3])
print("Przewidziane kolejne zdanie (najbliżej w naszym zestawie):")
print(sentences[best_idx])
```

```
Wymiary przewidzianego embedingu torch.Size([1, 1, 2048])
Początek historii:  Sentence[8]: "The moon reflected off the still lake."
Przewidziane kolejne zdanie (najbliżej w naszym zestawie):
A boy flew a kite on the open field.
```